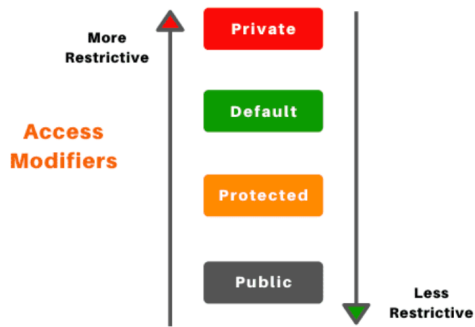
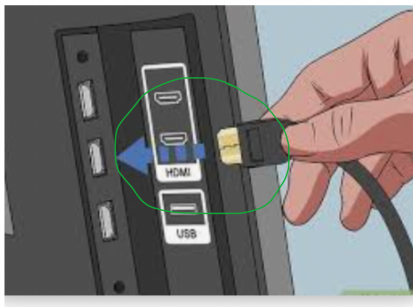


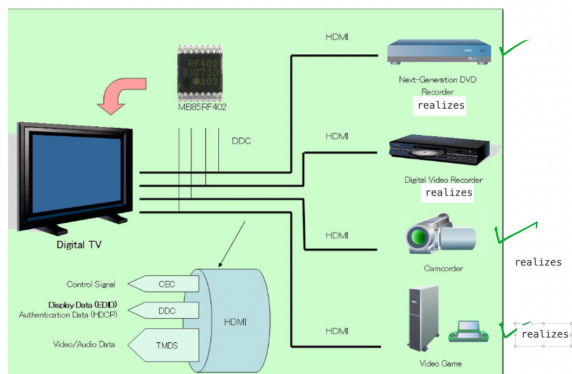


Access Modifier	Within Class	Within Package	Outside package without inheritance	Outside package with inheritance
Default	✓	✓	✗	✗
Private	✓	✗	✗	✗
Protected	✓	✓	✗	✓
Public	✓	✓	✓	✓

Realization Relationship:

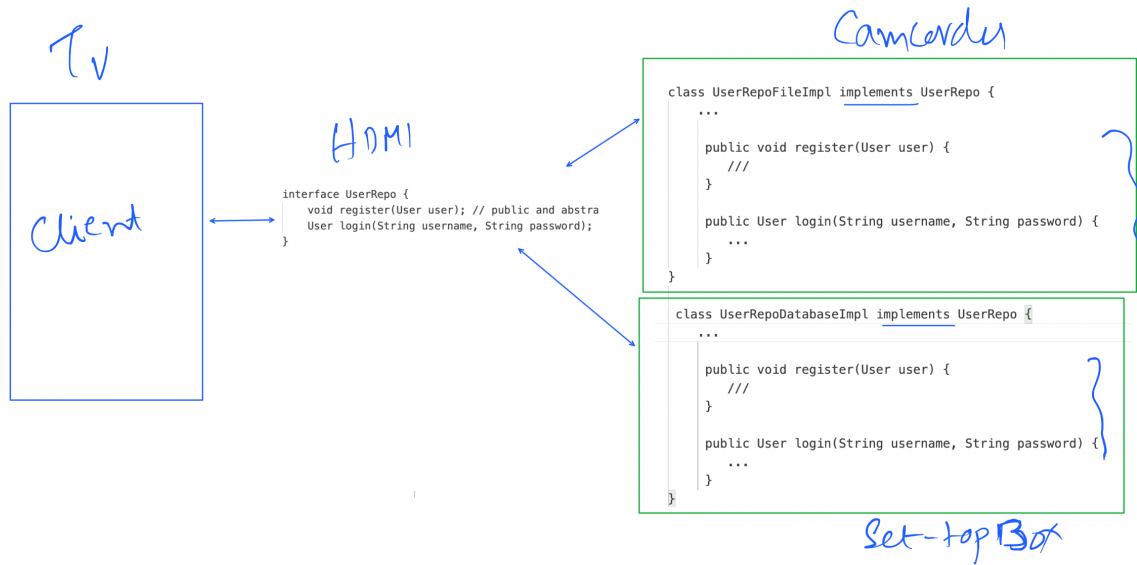


Contract

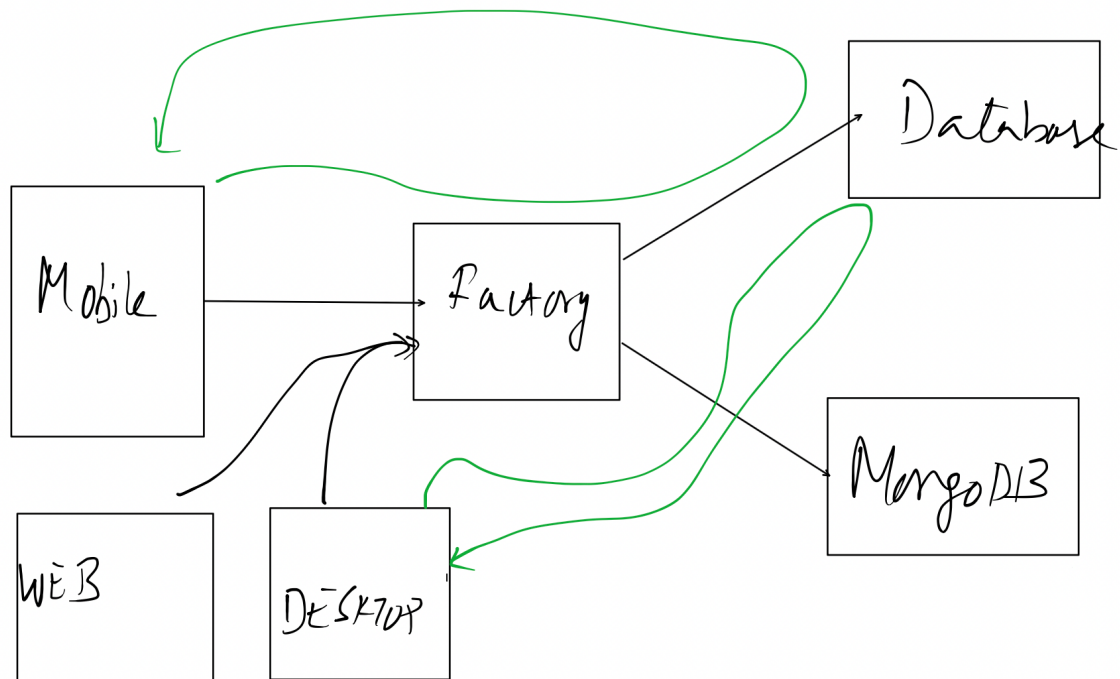


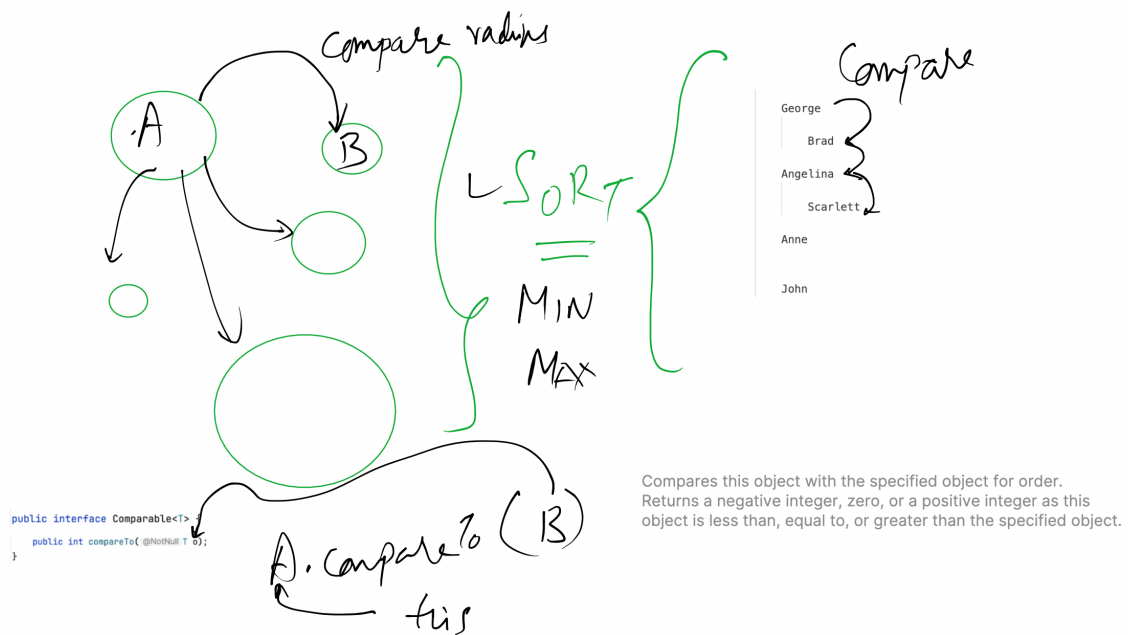
Interface





Object can realize many contracts / protocols





```
interface Swim {
    swim();
}

interface Dance {
    dance();
}

interface Fight {
    fight();
}
```

```
// actor is capable to dance
class Actor implements Dance {
    ...
    dance() {
    }
}

// hero is a actor; naturally hero also knows to dance
// hero is also capable to fight and swim
class Hero extends Actor implements Fight, Swim {
    fight() {...}
    swim() {...}
}
```

```
Fight f = new Fight(); // error, abstract

Fight f = new Actor(); // invalid; Actor doesn't know fighting

Fight f = new Hero(); // valid

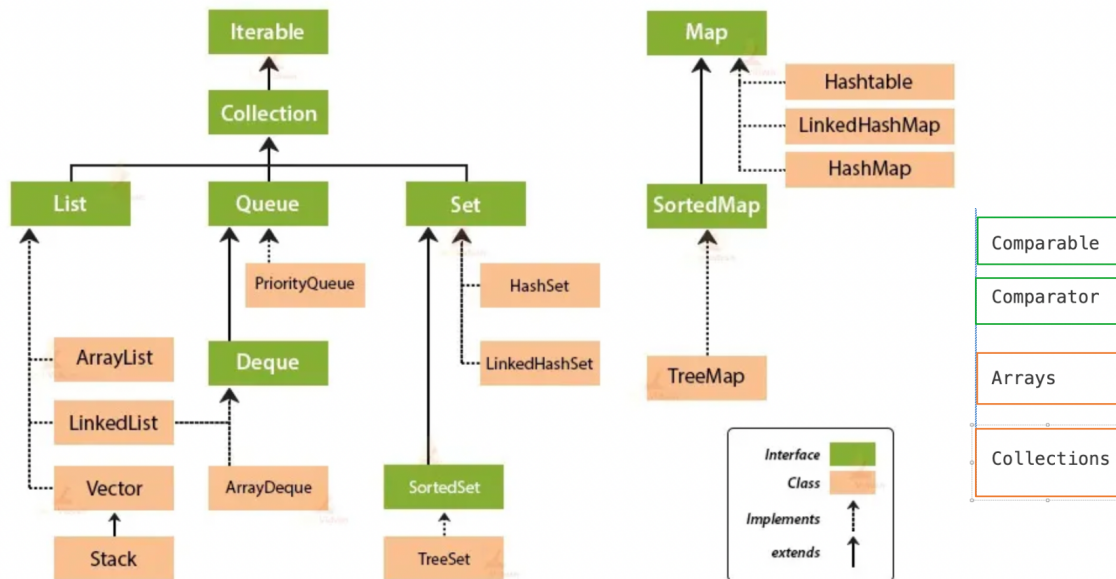
f.fight(); // works
f.dance(); // fails
f.swim(); // fails

Dance d = (Dance) f;
d.dance(); ✓
```

Stunt Master
DIRECTS FIGHT → SEQ ✓

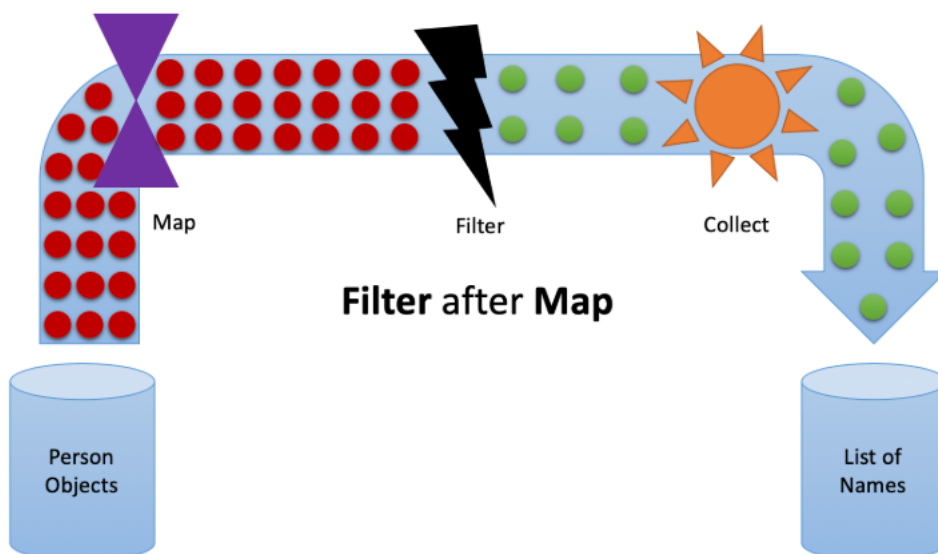
Stunt Master
Can't direct Dance

Collection Framework Hierarchy in Java



```
Arrays.sort(names, new Comparator<String>() {
    @Override
    public int compare(String o1, String o2) {
        return o1.length() - o2.length();
    }
});
```

Arrays.sort(names, (String o1, String o2) -> o1.length() - o2.length());



JAVA 8 STREAMS

